

PYONGYANG SUNAN INTL, Korea, Democratic People's Republic of

WMO: 470580

Lat: 39.224N

Lon: 125.670E

Elev: 117

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.7	(c) 9.2	(d) -8.6	(e) 3.5	(f) 11.2	(g) -4.9	(h) 4.3	(i) 14.3	(j) 19.0	(k) 19.3	(l) 16.3	(m) 19.5	(n) 3.8	(o) 320	(p) 0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.6	(c) 89.0	(d) 77.0	(e) 86.8	(f) 75.6	(g) 84.9	(h) 74.6	(i) 79.7	(j) 86.0	(k) 78.3	(l) 84.0	(m) 77.1	(n) 82.2	(o) 4.4	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.8	145.2	83.7	76.6	139.1	81.9	75.5	134.2	80.9	43.4	86.3	42.0	84.3	40.8	82.4	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.6	(b) 13.0	(c) 11.0	(e) 1.1	(f) 92.4	(g) 5.1	(h) 2.1	(i) -2.6	(j) 94.0	(k) -5.6	(l) 95.2	(m) -8.5	(n) 96.4	(o) -12.2	(p) 98.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			1.1	92.4	5.1	2.1	-2.6	94.0	-5.6	95.2	-8.5	96.4	-12.2	98.0		
			0.1	81.6	4.7	2.1	-3.3	83.1	-6.0	84.3	-8.6	85.5	-12.0	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	22.3	28.4	39.6	53.0	63.9	72.2	77.3	77.8	69.0	56.4	41.0	27.3
	DBStd	20.32	7.50	7.29	6.52	6.56	4.85	3.98	3.64	3.90	4.73	6.64	8.55	7.86
	HDD50	2848	857	606	327	43	0	0	0	0	0	23	287	705
	HDD65	5759	1322	1026	787	362	79	3	0	0	18	273	719	1170
	CDD50	3753	0	0	5	133	431	666	848	862	570	221	17	0
	CDD65	1190	0	0	0	2	45	219	383	397	138	6	0	0
	CDH74	8507	0	0	0	18	350	1322	2910	3241	643	23	0	0
	CDH80	2184	0	0	0	0	51	321	784	955	72	1	0	0
Wind	WSAvg	4.0	3.9	4.3	5.2	4.9	4.5	3.7	3.6	3.3	3.1	3.3	4.0	4.0
Precipitation	PrecAvg	39.8	0.4	0.7	0.9	2.0	3.3	4.1	12.6	8.6	4.1	1.9	1.6	0.7
	PrecMax	66.0	1.1	3.0	2.6	4.2	8.5	10.7	32.5	24.4	13.1	4.9	4.7	1.9
	PrecMin	23.0	0.0	0.0	0.0	0.0	0.5	1.0	3.5	1.1	0.6	0.0	0.2	0.1
	PrecStd	10.7	0.3	0.7	0.6	1.1	2.0	2.2	5.9	5.8	3.6	1.1	1.2	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	52.0	64.3	76.6	84.9	89.2	92.2	91.7	85.0	76.9	64.7	49.1
		MCWB	38.4	44.1	52.1	59.2	64.7	70.9	78.9	79.4	72.9	63.7	55.9	44.0
	2%	DB	39.3	46.4	58.5	72.1	80.7	86.1	88.7	89.4	82.1	72.7	60.0	44.6
		MCWB	35.0	39.6	47.4	56.4	63.6	69.8	77.0	78.8	70.6	62.2	53.4	40.8
	5%	DB	36.1	42.7	54.7	68.8	77.9	83.6	86.3	87.3	80.0	69.7	56.8	41.4
		MCWB	32.5	36.4	45.2	54.9	61.7	69.2	76.4	77.5	68.8	60.0	51.2	37.9
	10%	DB	33.4	39.4	51.0	65.4	75.0	81.2	84.3	85.3	77.9	66.8	53.6	38.4
		MCWB	30.5	34.5	42.7	52.8	60.7	68.0	75.6	76.4	67.8	58.5	48.5	35.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	45.6	53.8	63.8	69.1	75.4	81.5	82.2	76.1	66.3	58.2	45.2
		MCDB	42.3	48.9	61.2	71.4	78.1	83.2	89.2	88.2	81.1	72.8	62.0	47.7
	2%	WB	35.7	40.4	49.6	59.4	66.3	73.2	79.3	80.2	73.7	63.7	54.6	41.4
		MCDB	38.7	45.8	56.4	69.1	76.2	81.2	85.3	86.6	78.6	69.8	58.9	44.3
	5%	WB	33.1	37.5	46.3	56.6	64.4	71.6	78.2	78.9	71.7	61.9	51.9	38.2
		MCDB	35.5	41.4	53.6	65.3	73.9	78.9	83.5	85.2	77.1	67.7	56.2	41.0
	10%	WB	30.8	34.9	43.5	54.3	62.5	70.3	77.1	77.7	70.0	59.9	49.1	35.3
		MCDB	33.2	38.8	49.7	63.1	70.9	77.3	82.1	83.2	75.8	65.7	52.8	37.9
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.2	16.6	18.8	18.9	16.1	11.7	12.6	16.2	17.5	14.4	12.7
		MCDBR	16.0	19.2	22.5	24.2	23.8	20.7	15.1	14.8	17.7	19.6	17.8	15.2
		MCWBR	12.4	13.0	12.9	11.3	9.3	8.0	6.7	6.6	7.6	9.8	11.7	11.8
	5% WB	MCDBR	14.5	16.8	21.0	20.3	19.6	15.7	11.9	12.7	13.8	17.0	16.1	14.3
		MCWBR	12.1	12.6	13.7	11.3	9.4	7.7	6.3	6.4	7.3	10.0	12.0	11.9
Clear-Sky Solar Irradiance	taub	0.419	0.494	0.554	0.630	0.647	0.728	0.704	0.594	0.519	0.541	0.503	0.431	
	taud	1.930	1.796	1.708	1.612	1.625	1.536	1.626	1.848	1.954	1.802	1.833	1.931	
	Ebn at Noon	220	219	221	215	215	198	201	221	230	202	189	203	
	Edh at Noon	43	57	69	81	81	89	80	63	53	56	48	40	
All-Sky Solar Radiation	RadAvg	768	1013	1382	1616	1808	1671	1327	1418	1391	1103	741	655	
	RadStd	56	74	106	125	163	170	173	166	118	71	75	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+83

Nomenclature: See separate page